Leps-P: A Probabilistic and Updatable Extension of the Leps Shear Strength Envelopes for Rockfill and Mine Waste Rock

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Abstract

The Leps envelopes, recently updated with a database of more than 900 large-scale triaxial tests (Leps 2.0), are the standard empirical estimate of the drained shear strength of rockfill and coarse mine waste rock. Waste dumps 500–700 m high and reliability-based design require more than a central envelope: they need a strength distribution that depends on stress, can be updated with site tests, and states how much uncertainty is added when extrapolating beyond the data. This paper presents Leps-P, a three-level hierarchical model of the Leps power envelope in which the reference secant friction angle and the curvature exponent vary randomly between laboratories and between materials, the within-material dispersion depends on stress, and hyperparameter uncertainty is propagated by Markov chain Monte Carlo sampling. The model was calibrated with 767 tests on 226 materials from 36 publications and evaluated by leaving out one publication at a time. Without site tests, the 80% predictive interval contained 83% of the held-out tests, with 9% below the 10th percentile; its accuracy was equivalent to that of the deterministic envelope with constant dispersion. With site tests at low stress and an updating rule that guards against prior–data conflict, the continuous ranked probability score at high stress improved significantly (0.041 to 0.030). Between-laboratory differences accounted for 37% of the between-material variance of the reference angle. Within-material comparisons showed a specimen-size effect consistent with Weibull scaling but about 60% of its usual magnitude ($\kappa = 0.24 \pm 0.11$). Applied to waste rock from a polymetallic skarn deposit, the strength uncertainty alone gave negligible probabilities of failure for dumps up to 700 m, whereas a saturated,

contractive base controlled stability. Leps-P, its code, and its parameters can be recalibrated as high-pressure tests become available.

Practical Applications

Engineers who design high waste dumps and rockfill dams often have few or no large-scale shear tests of their own and rely on published charts. This paper turns the most widely used chart into a tool that gives not only a best estimate of strength but also its range, how that range grows at the very high stresses found at the base of tall dumps, and how much it shrinks when a few laboratory tests of the site material are added. The method also warns the user when site tests disagree with the published experience, and when strength is being extrapolated beyond the stresses that have ever been tested. An example for a 700 m waste dump shows that, once the strength of the waste rock is characterized, the safety of a very high dump depends mostly on keeping its base drained: a saturated base that cannot drain during construction can reduce the factor of safety below one even when the drained strength is well known. The free software that accompanies the paper can be used to obtain design strength envelopes and to update them as new tests become available.

Author keywords: Rockfill; Mine waste rock; Shear strength; Nonlinear failure envelope; Hierarchical Bayesian model; Reliability-based design; High waste dumps; Specimen-size effect.

Introduction

Leps (1970) compiled 109 large-scale triaxial tests and proposed three nonlinear envelopes for rockfill. For five decades those curves were used for dams and waste dumps whenever site-specific tests were unavailable. Cylwik et al. (2025) updated them with a public database of 938 tests (Arrieta 2025), adopted the power criterion of De Mello (1977) and Charles

and Watts (1980) with a single fitting parameter, the secant friction angle at $\sigma_n' = 10p_a$, and related that angle to rock quality and void ratio. That update is referred to as Leps 2.0.

Two developments in mining practice call for a further step. First, waste dumps now reach 500–700 m (Valenzuela et al. 2008), so effective normal stresses at their base reach 10–15 MPa in a material that crushes, loses dilatancy and permeability, and may respond contractively (Linero et al. 2007; Ovalle et al. 2020), whereas the largest normal stress in the database is 7.9 MPa. Second, design is moving toward reliability-based criteria (Read and Stacey 2009; Hawley and Cunning 2017), which require a strength distribution, its dependence on stress, and a rule to combine published experience with site tests.

Bayesian and hierarchical models have been used to characterize geotechnical properties from generic databases and few site data (Ching and Phoon 2019; Ching et al. 2021) and, for rockfill, to infer the parameters of the Barton and Kjærnsli (1981) criterion for material groups of a single project (Contreras et al. 2022; Contreras and Linero-Molina 2025). The present work builds on both lines and contributes: (1) a hierarchical model of the stress-dependent envelope itself, with random reference angle and curvature and physical covariates, calibrated on the worldwide database; (2) separation of laboratory (publication) and material variability; (3) systematic out-of-sample evaluation by leaving out entire publications; (4) an updating rule with a prior–data conflict diagnostic, validated out of sample; (5) quantification of shape and specimen-size uncertainty in extrapolation; and (6) a design format linked to characteristic values and probability of failure. The approach is illustrated with waste rock from a polymetallic skarn deposit in the Andes and with waste dumps 150–700 m high.

Database and Model Formulation

Calibration Database

The public database of Leps 2.0 (Arrieta 2025) was used. Tests with $\sigma_3' \geq 60$ kPa, reported initial void ratio, and at least two tests per set were retained: 767 tests on 226 sets from 36 publications. A set is a material with a given gradation, density, and lithology tested in one program. Rock quality was assigned following Table 5 of Cylwik et al. (2025), and failure was taken at maximum deviator stress, as in the database. Fig. 1 shows that the data are concentrated between 0.1 and 3 MPa; 90 tests exceed 3 MPa and come from eight publications.

Hierarchical Envelope

The Leps 2.0 criterion is the power law

$$\tau = \tan\phi_1' \sigma_0^{1-m} {\sigma_n'}^m \tag{1}$$

where $\sigma_0 = 10p_a$, $p_a = 0.1013$ MPa is atmospheric pressure, and ϕ_1' is the secant friction angle at $\sigma_n' = \sigma_0$. In logarithmic variables $x = \ln(\sigma_n'/\sigma_0)$ and $y = \ln(\tau/\sigma_0)$, Eq. (1) is a straight line with intercept $\theta = \ln\tan\phi_1'$ and slope m. Leps-P treats every set j tested in publication k as a realization of a population:

$$y_{ijk} = \theta_{jk} + m_{jk}x_{ijk} + c_2 x_{ijk}^2 + \varepsilon_{ijk} \tag{2}$$

$$\theta_{jk} = \beta_{c(jk)} + \beta_e e_{jk} + b_{0k}^{(p)} + b_{0jk}^{(s)}, m_{jk} = \beta_x + b_{1k}^{(p)} + b_{1jk}^{(s)} \tag{3}$$

$$\mathbf{b}^{(p)} \sim N(\mathbf{0}, \mathbf{G}_p), \mathbf{b}^{(s)} \sim N(\mathbf{0}, \mathbf{G}_s), \ln\sigma_\varepsilon(x) = a_0 + a_1 x \tag{4}$$

where β_c is the intercept of rock-quality class c (good, moderate, or low), β_e is the effect of initial void ratio e, β_x is the mean curvature, $c2$ is a population term that lets the local slope change with stress, $\mathbf{b}^{(p)}$ and $\mathbf{b}^{(s)}$ are the publication (laboratory and procedure) and set (material) random effects on (θ, m) with covariance matrices $\mathbf{G}_p$ and $\mathbf{G}_s$, and ε is the dispersion between specimens of the same material. Eqs. (2)–(4) form a linear mixed model (Laird and

Ware 1982) with heteroscedastic residuals. With β_e and β_c, the median of ϕ_1' approximates the relation $\phi_1' = \phi_{e=0}' - \omega e$ of Leps 2.0; with $\mathbf{G}_p = \mathbf{G}_s = 0$, $\sigma_\varepsilon = 0$ and $c_2 = 0$, Eq. (2) reduces to Eq. (1).

Predictive Distribution and Shape Constraint

For a new material of class c and void ratio e tested by a new laboratory, y at stress x is normal with

$$E[y] = \beta_c + \beta_e e + \beta_x x + c_2 x^2, \mathrm{Var}[y] = \mathbf{z}^T(\mathbf{G}_p + \mathbf{G}_s)\mathbf{z} + \tilde{\mathbf{x}}^T \mathbf{C}_\beta \tilde{\mathbf{x}} + \sigma_\varepsilon^2(x) \tag{5}$$

with $\mathbf{z} = (1, x)^{\mathrm{T}}$, $\tilde{\mathbf{x}}$ the covariate vector, and $\mathbf{C}_\beta$ the covariance of the fixed effects. The first term is the uncertainty of the mean envelope of the material; the last is the scatter of a single test about it. The distinction matters in design: a long failure surface averages the second term but not the first (Vanmarcke 1977; Christian et al. 1994). Hyperparameter uncertainty is propagated by sampling ($\mathbf{G}_p$, $\mathbf{G}_s$, a_0, a_1) from their posterior and, for each sample, the fixed effects from N($\widehat{\boldsymbol{\beta}}$, $\mathbf{C}_\beta$). A power envelope whose local slope exceeds one is convex and its secant angle grows without bound, which has no physical support in granular materials; samples whose local slope $m + 2c_2x$ exceeds one for $\sigma_n' \leq 20$ MPa are therefore discarded.

Updating with Site Tests and Prior–Data Conflict

Given n site tests (x_i, y_i) of one material, the random effects $\mathbf{b} = \mathbf{b}^{(p)} + \mathbf{b}^{(s)}$ are updated with the exact Gaussian posterior

$$\mathbf{C}_{post} = (\mathbf{G}^{-1} + \mathbf{Z}^T \mathbf{R}^{-1} \mathbf{Z})^{-1}, \boldsymbol{\mu}_{post} = \mathbf{C}_{post} \mathbf{Z}^T \mathbf{R}^{-1} \mathbf{r} \tag{6}$$

where $\mathbf{G} = \mathbf{G}_p + \mathbf{G}_s$, $\mathbf{Z}$ has rows $(1, x_i)$, $\mathbf{R} = \mathrm{diag}\sigma_\varepsilon^2(x_i)$, and $\mathbf{r}$ is the vector of residuals with respect to the prior mean. Triaxial results are converted to (σ_n', τ) with Eqs. (13), (14), and (16) of Cylwik et al. (2025). Before updating, consistency between site tests and prior is checked with the prior predictive Mahalanobis distance

$$D^2 = \mathbf{r}^T \left(\mathbf{Z}\mathbf{G}\mathbf{Z}^T + \tilde{\mathbf{X}}\mathbf{C}_\beta \tilde{\mathbf{X}}^T + \mathbf{R} \right)^{-1} \mathbf{r} \sim \chi^2_n \quad (7)$$

Updating both θ and m requires tests that span a range of stresses. The rule adopted, and validated below, is: update (θ, m) jointly only if the site tests span a ratio of normal stresses of at least three and the conflict test is passed ($p \geq 0.05$); otherwise update only the level of the envelope at the centroid of the site data, $\overline{x}$, and keep the prior conditional distribution of the slope. In the second case the envelope pivots about the tested range instead of about σ_0, which avoids spurious extrapolation toward low stress.

Specimen Size

Following the Weibull scaling of Frossard et al. (2012) and Ovalle et al. (2014), the envelope of a material with maximum particle size D_f is obtained from tests at size D_l by a shift in θ with m unchanged:

$$\Delta\theta = \beta_D \ln\left(D_f / D_l\right), \beta_D = -\kappa(1-m) \quad (8)$$

where κ is the ratio of a size exponent to the Weibull modulus. Rather than adopting κ from particle-crushing tests, β_D was estimated within materials of the database tested at several sizes (see Results), and it enters Leps-P as a random variable with the between-material dispersion found there.

Estimation and Out-of-Sample Evaluation

The fixed effects and random effects were integrated analytically and the variance parameters were estimated by restricted maximum likelihood with publication blocks. The posterior of the eight variance parameters was sampled with an adaptive Metropolis algorithm on the restricted likelihood (two chains; potential scale reduction ≤ 1.04). Two forms were fitted: the quadratic form of Eq. (2) and a pure power form ($c_2 = 0$).

The model was refitted leaving out each publication and its tests were predicted, so every prediction refers to a laboratory, material, and procedure not used in calibration. To check that the class assignment, which Cylwik et al. (2025) based on measured strength, does not leak information, the evaluation was repeated leaving out each material type. Two uses were evaluated. In Mode A (no site tests) the predictive distribution of Eq. (5) was scored. In Mode B, each held-out set with at least two tests at $\sigma_n' \leq 1.5$ MPa and one at $\sigma_n' \geq 2$ MPa was updated with the former, and the latter were predicted. Metrics were the coverage of the central 80% interval, the fraction of tests below the 10th percentile (which should be close to 10% for a design envelope at that percentile to mean what it says), the mean width of the 80% interval, the continuous ranked probability score (CRPS; Gneiting and Raftery 2007), and the quantile (pinball) loss at 10%. Confidence intervals were obtained by bootstrapping whole publications. The reference was the Leps 2.0 envelope ($m = 0.9$) with a constant dispersion in ln τ estimated in the same training fold.

Results

Parameters

Table 1 lists the parameters. The mean curvature is $\beta_x = 0.860$ and the quadratic term is $c_2 = 0.006 \pm 0.002$, so the local slope rises with stress, from about 0.84 at 0.2 MPa to about 0.89 at 15 MPa. This agrees with the increase of the power exponent with stress range reported by Charles and Watts (1980) and Indraratna et al. (1993). The reference angles at $e = 0$ are 50.42°, 47.67°, and 45.67° for good, moderate, and low rock quality, and the void-ratio effect is about 15° per unit of e at $\phi_1' \approx 41°$, compared with 20° in Leps 2.0. The median ϕ_1' of a new material reproduces the Leps 2.0 table within 1.8° for $0.3 \leq e \leq 0.6$; because Leps-P estimates the

curvature (0.86) instead of fixing it at 0.9, its median envelope is lower than the Leps 2.0 envelope away from the anchor, by about 2° at 10 MPa for moderate rock quality.

Heteroscedasticity was strongly supported: the within-material dispersion decreases from about 0.056 at 0.2 MPa to 0.036 at σ_0 and 0.020 at 7.5 MPa (Table 1). The publication level was also strongly supported: laboratories and procedures account for 37% of the between-material variance of θ and for most of that of m. At σ_0, between-material and between-laboratory differences make up 89% of the predictive variance. Within-material dispersion, which is mostly specimen-to-specimen and testing error, is a small part of the total.

Calibration without Site Tests

Leaving out each publication, the 80% interval contained 83% of the tests (95% CI 78%–88%) and 9% fell below the 10th percentile (Fig. 3, Table 2). Leaving out each material type gave 83% and 8%, so the class assignment does not inflate the scores. Replacing the class by the uniaxial compressive strength of the rock degraded all metrics (CRPS 0.085 versus 0.069). The CRPS of Leps-P and of the reference were statistically indistinguishable (difference +0.0008, 95% CI −0.0048 to +0.0075). Without site tests, therefore, Leps-P does not predict the strength of a new material better than Leps 2.0 with a constant dispersion; its contribution in that case is a calibrated and stress-dependent description of the uncertainty.

Above 2.75 MPa the Leps-P intervals were wider than necessary: coverage was 88%, only 4% of the tests fell below the 10th percentile, and the 10% quantile loss favored the reference. Materials converge at high stress (Indraratna et al. 1993) faster than the random-slope model assumes. Leps-P is thus conservative at high stress, and the extra width is the price of not assuming the convergence beyond the data.

Value of Site Tests

In Mode B (46 sets, 64 tests at $\sigma_n' \geq 2$ MPa from 13 publications), the prior intervals were too wide for these materials (coverage 94%). Updating θ and *m* jointly narrowed the interval but made it overconfident (standard deviation of standardized residuals 1.22) and did not improve the CRPS significantly. With the updating rule, the CRPS improved from 0.041 to 0.030 (difference +0.0105, 95% CI +0.0053 to +0.0155), the mean width of the 80% interval fell from 0.256 to 0.176 in ln τ, coverage was 91%, and 3% of the tests fell below the 10th percentile (Fig. 4). A few conventional tests at low stress thus improve both the accuracy and the sharpness of the strength predicted at high stress. The conflict test flagged 7% of the sets.

Form of the Envelope in Extrapolation

The quadratic and power forms are indistinguishable within the data (Fig. 2) but diverge beyond them: for moderate rock quality and $e = 0.45$, the median secant angle at 15 MPa is 31.6° with the quadratic form and 30.5° with the power form, and the 10th percentiles are 27.2° and 26.0°. The power form is the conservative choice when extrapolating. A floor at a critical-state angle was not imposed because the critical-state line of crushable rockfill moves with breakage (Ventini et al. 2021) and the critical-state angle itself decreases with stress and breakage in large triaxial tests (Xiao et al. 2014a, b); such a floor cannot be identified as a material constant with the present data.

Specimen-Size Effect

Eighteen materials of the database were tested at more than one maximum particle size (Fig. 5). Across all of them the mean β_D was -0.009 ± 0.017 with a large between-material dispersion (0.065), and the sign of the effect varied. In most groups with a positive β_D the gradation became wider or the density increased as D_{max} increased, so gradation and packing,

not size alone, controlled the result (see also Alonso et al. 2012). In the five groups with parallel gradation and similar density, $\beta_D = -0.041 \pm 0.015$ ($\kappa = 0.24 \pm 0.11$), consistent in sign with Weibull scaling (Frossard et al. 2012) but about 60% of the magnitude implied by $\kappa \approx 0.4$, in line with the observation of Ovalle et al. (2014) that the method is conservative. For a material tested with D_{max} = 178 mm and placed with $D_{max} \approx 1$ m, the median reduction in ϕ'_s is about 2° with a standard deviation of about 1.5°. The scale shift is therefore a source of uncertainty of the same order as the between-material variability, and it is reduced only by large-scale tests of the site material.

Application: Waste Rock from a Polymetallic Skarn Deposit

Site Data

The deposit is a polymetallic skarn in the Andes whose waste rock is composed of metamorphic carbonate rock (marble), hornfels, intrusive rock, and sedimentary carbonate rocks (limestone and marl). Drained triaxial tests on specimens 0.6–1.0 m in diameter without tailings were used: 17 tests in five sets, D_{max} = 178 mm, σ'_3 = 0.5–2.5 MPa, and failure at maximum deviator stress. Replicates of one material were grouped, and single tests on smaller specimens were excluded. Rock quality was assigned as low for the metamorphic carbonate, intrusive, and sedimentary carbonate rocks, and as moderate for hornfels. The sedimentary carbonate rocks have no tests and keep the prior. Illustrative mass fractions of 40%, 20%, 15%, and 25% were used for the dump.

Envelopes by Lithology

Table 3 and Fig. 6 give the field-scale envelopes, after the size shift from 178 mm to 1,000 mm. The metamorphic carbonate and intrusive sets spanned stress ratios of about four and passed the conflict test, so θ and m were updated. The hornfels tests spanned a ratio of 2.6 and

failed the conflict test ($p < 0.001$): their secant angle increased from 41.7° to 45.1° as σ_n' rose from 1.6 to 4.2 MPa, which the population model regards as implausible. The level of the hornfels envelope was updated at the centroid of the tests, and its slope kept the prior. The likely cause is the failure definition: the specimens were still contracting and hardening at 11–20% axial strain, so the maximum deviator reflects the strain limit of the test. The diagnostic thus identifies data that should be reviewed before they are trusted in extrapolation. The metamorphic carbonate, which dominates the dump, loses strength with stress faster than the intrusive rock, while the untested sedimentary rocks have the widest band and are the largest source of uncertainty.

Dump-Average Envelope and Correlation between Lithologies

A failure surface through a mixed dump mobilizes the mass-weighted strength of the lithologies it crosses. The lithology envelopes were sampled jointly, sharing hyperparameters, fixed effects, and size shift (systematic uncertainty), with independent material effects. Three assumptions bound the result at 15 MPa (Table 3): the model correlation gives a P05 of 30.7°, full correlation between lithologies gives 29.6°, and a surface that crosses only the weakest lithology gives 23.8°. The difference between the last two values, about 6°, is larger than any other source of uncertainty considered and depends on how the lithologies are placed in the dump, which is a design decision.

Stability of Dumps 150–700 m High

Dumps with an overall slope of 2H:1V, unit weight 21 kN/m³, and a competent level foundation were analyzed with the envelopes as tabulated functions, using Bishop's (1955) method with a circular search in a purpose-written code and the Morgenstern and Price (1965) and Spencer (1967) methods in Slide2 9.043 (Rocscience year). The two programs agreed within

0.010; the purpose-written code was slightly higher. Table 4 summarizes the results. For the 700 m dump, the factor of safety is 1.65 with the median envelope of the site waste rock, 1.50 with its characteristic (P05) envelope, 1.28 if the surface crosses only the weakest lithology, and 1.28 for a generic material of moderate quality without tests. Even for 700 m, the critical circle reached $\sigma_n' \approx 2.5$ MPa, within the calibration range: on a competent level foundation the global factor of safety of a high dump is controlled by stresses of 1–3 MPa, not by the extrapolated part of the envelope.

A Monte Carlo analysis with 400 joint samples of the dump-average envelope gave, for the 700 m dump, a median factor of safety of 1.67, a 10th percentile of 1.56, a minimum of 1.46, and a coefficient of variation of 0.054, with no failures. The factor of safety computed with the P10 envelope approximated the 10th percentile of the factor of safety. The uncertainty of the drained strength alone, once quantified, does not control the reliability of a high dump on a competent, drained foundation.

Base Saturation and Undrained Response

The 700 m dump was also analyzed on a 10° foundation slope with a 30 m basal layer, using a nonlinear search optimized by simulated annealing and the Morgenstern–Price method (Table 5). Dry, the critical surface remained nearly circular within the waste rock (factor of safety 1.65 and 1.50 with the P50 and P05 envelopes). With a saturated basal layer and an excess pore-pressure ratio of 0.2, it dropped to 1.53 and 1.39. With an undrained basal layer whose strength is proportional to the vertical effective stress, $S_u = 0.17\sigma_v'$ (the isotropic triaxial ratio of 0.28 reported by Valenzuela et al. (2008) referred to σ_v' under $K_0 \approx 0.4$) and $S_u = 0.09\sigma_v'$ (the mean liquefied strength ratio of Olson and Stark (2002), extrapolated from much lower stresses), the factor of safety was 0.65 and 0.39.

Excess pore pressure from consolidation was estimated with one-dimensional consolidation under time-dependent loading (Olson 1977) for a constrained modulus of 50 MPa and a drained top. For the slow overall growth of the dump it is negligible unless $k \leq 10^{-8}$ m/s, but a dumping front that adds 150 m in three months produces $\Delta u/\sigma_v = 0.11$ in a 30 m basal layer with $k = 10^{-7}$ m/s and 0.77 in a 100 m layer. The rate of advance of the dumping front and the permeability of the crushed or degraded base are therefore first-order design variables, as in the liquefaction flowslides of coal-mine waste dumps (Dawson et al. 1998; Hungr et al. 2002).

Discussion

What Leps-P Adds

Leps-P does not predict the central strength of an untested material better than Leps 2.0; its value lies in uncertainty that is calibrated and stress dependent, and in making site tests count. Because most of the predictive variance lies between materials and laboratories, it is not reduced by averaging along a failure surface within one material, but by testing that material at large scale over a stress range wide enough to update the curvature.

Design Format

Following Eurocode 7 as interpreted by Prästings et al. (2019), the characteristic value for a global failure mechanism is a cautious estimate of the mean strength along the surface. In Leps-P it is the 5th percentile of the mean envelope (excluding within-material scatter), with the systematic components (fixed effects, laboratory, material, shape, and size) retained. Reliability is then checked with the probability of failure computed by sampling the envelope, with correlations between lithologies stated explicitly. Applying a global factor of safety to a P05 envelope counts the strength uncertainty twice; the target factor of safety should be calibrated to

the consequence class and to the remaining sources of uncertainty, chiefly pore pressure and foundation conditions (Christian et al. 1994; Read and Stacey 2009).

Limitations

The calibration database contains no tests above σ_n' = 7.9 MPa, and its high-stress tests come from eight publications on basalt, gravel, argillite, diorite, and conglomerate; carbonate and argillaceous waste rock are not represented at high stress. Above 7.9 MPa the band is an extrapolation of the model. The class assignment for lithologies not listed by Cylwik et al. (2025) is a judgment. Specimens were scalped or of parallel gradation with $D_{max} \leq 200$ mm, and the relation with field material depends on gradation as well as size (Girumugisha et al. 2024). The within-material residual is laboratory scatter and does not represent spatial variability in the dump, which was not modeled. Leps-P describes drained strength; undrained and liquefied strengths of crushed waste rock at high stress were taken from lower-stress studies and must be measured. Leps-P is defined by its population parameters and by the procedure to estimate and evaluate them, so these gaps can be closed by re-estimation as data appear: drained large-scale tests at σ_3' of 5–10 MPa on carbonate and argillaceous waste rock, undrained tests at high pressure on crushed gradations, and tests of one material at two specimen sizes with parallel gradation and equal density would most reduce the uncertainty of high dumps.

Conclusions

A three-level hierarchical extension of the Leps power envelope, Leps-P, was calibrated with the Leps 2.0 database and evaluated out of sample. The main conclusions are:

1. Without site tests, the Leps-P 80% interval covered 83% of held-out tests and its accuracy equaled that of Leps 2.0 with constant dispersion; the intervals were conservative above 2.75 MPa.

2. With a few site tests at low stress and the proposed updating rule, the CRPS at high stress improved significantly, from 0.041 to 0.030, while calibration was retained.
3. Laboratories and procedures explained 37% of the between-material variance of the reference angle; within-material scatter was a small part of the predictive variance.
4. Within materials of parallel gradation, the specimen-size effect agreed in sign with Weibull scaling but had about 60% of its magnitude ($\kappa = 0.24$); size enters design as an uncertainty of about $2° \pm 1.5°$ from laboratory to field.
5. For dumps up to 700 m on a competent drained foundation, the drained strength uncertainty gave negligible probability of failure; segregation of the weakest lithology, base saturation, and undrained response controlled stability.

Data Availability Statement

Some or all data, models, or code generated or used during the study are available in a repository online in accordance with funder data retention policies (the Leps-P code, calibrated parameters, and evaluation scripts: repository DOI to be assigned; calibration database: Arrieta 2025). Some or all data, models, or code generated or used during the study are proprietary or confidential in nature and may only be provided with restrictions (the test results of the skarn deposit, which belong to a third party and are published in anonymized and aggregated form with its consent to be confirmed).

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Figure Captions List

Fig. 1. Distribution of effective normal stress on the failure plane in the calibration database, compared with the stresses in a 700 m waste dump.

Fig. 2. Predictive distribution of the secant friction angle of a new material of moderate rock quality with e = 0.45 tested by a new laboratory: 80% interval, median, and 10th percentile for the quadratic and power forms. The shaded zone exceeds the largest normal stress in the database.

Fig. 3. Out-of-sample calibration without site tests, leaving out each publication: (a) coverage of the 80% interval; and (b) fraction of tests below the 10th percentile, by range of normal stress. Dashed lines are the nominal values.

Fig. 4. Mean continuous ranked probability score of ln τ for held-out tests at $\sigma_n' \geq 2$ MPa, without and with updating from site tests at $\sigma_n' \leq 1.5$ MPa. Thin lines are publications; the thick line is the mean of the 64 tests.

Fig. 5. Within-material specimen-size coefficient β_D for materials tested at more than one maximum particle size (error bars: 95% confidence intervals). The shaded band is the 95% confidence interval of the mean for parallel gradation and equal density; the dotted line is Weibull scaling with $\kappa = 0.41$.

Fig. 6. Field-scale envelopes of the waste rock of the skarn deposit by lithology (median and 10th percentile) and dump-average envelope. The shaded zone is beyond the stresses tested at the site.

Table 1. Parameters of Leps-P (quadratic form)

Parameter	Estimate	Standard error or 90% credible interval
β_c, good rock quality	0.190	0.033 ($\phi'_{e=0}$ = 50.4°)
β_c, moderate rock quality	0.093	0.028 ($\phi'_{e=0}$ = 47.7°)
β_c, low rock quality	0.024	0.024 ($\phi'_{e=0}$ = 45.7°)
β_e	−0.523	0.049
β_x	0.860	0.008
c_2	0.006	0.002
SD of θ, publication	0.063	0.046–0.088
SD of m, publication	0.042	0.029–0.059
Correlation of θ and m, publication	−0.49	−0.80 to −0.07
SD of θ, set	0.081	0.074–0.090
SD of m, set	0.033	0.027–0.040
Correlation of θ and m, set	−0.44	−0.62 to −0.24
σ_ε at $\sigma'_n = \sigma_0$	0.036	0.034–0.038
a_1	−0.297	−0.359 to −0.230

Note: 767 tests, 226 sets, 36 publications. Fixed effects: restricted maximum likelihood estimate and standard error; variance parameters: posterior median and 90% credible interval.

Table 2. Out-of-sample evaluation

Evaluation	Tests	80% coverage	Below P10	CRPS (Leps-P)	CRPS (reference)
Mode A, leave one publication out, all	767	83%	9%	0.0688	0.0696
Mode A, leave one publication out, σ'_n > 2.75 MPa	100	88%	4%	0.0468	0.0446
Mode A, leave one material type out, all	767	83%	8%	0.0685	0.0698
Mode B, without site tests	64	94%	3%	0.0410	—
Mode B, joint update of θ and m	64	88%	3%	0.0371	—
Mode B, updating rule	64	91%	3%	0.0305	—

Note: Nominal values: 80% coverage and 10% below P10. Reference: Leps 2.0 (m = 0.9) with constant dispersion. Mode A rows for leave one material type out use the power form.

Table 3. Field-scale secant friction angle (°) of the skarn waste rock by lithology and for the dump

Material	ϕ_1' / m (lab)	Update	Conflict p	1 MPa	3 MPa	10 MPa	15 MPa
Metamorphic carbonate	39.1° / 0.93	θ and m	0.171	37.2 / 35.3	35.1 / 33.3	33.2 / 31.1	32.6 / 30.4
Hornfels	48.6° / 0.87	level only	<0.001	46.6 / 43.8	42.6 / 40.7	38.6 / 35.8	37.4 / 34.0
Intrusive	39.9° / 0.96	θ and m	0.273	37.9 / 36.1	36.9 / 35.1	36.1 / 34.1	35.8 / 33.8
Sedimentary carbonate (untested)	40.7° / 0.86	prior	—	38.8 / 34.7	34.6 / 30.9	30.6 / 26.5	29.3 / 25.0
Dump average, model correlation				39.8 / 37.1	36.9 / 34.4	34.1 / 31.6	33.3 / 30.7
Dump average, full correlation				39.8 / 36.4	36.9 / 34.0	34.1 / 30.7	33.3 / 29.6
Weakest lithology only				38.8 / 33.5	34.6 / 30.0	30.6 / 25.4	29.3 / 23.8

Note: Lithology rows give P50 / P10 of the mean envelope; dump rows give P50 / P05. Size shift from 178 to 1,000 mm. No site tests above $\sigma_n' \approx 4.3$ MPa.

Table 4. Factor of safety of dumps on a competent level foundation (Morgenstern–Price method)

Envelope	150 m	400 m	700 m
Site waste rock, P50	1.93	1.74	1.65
Site waste rock, P05	1.73	1.58	1.50
Weakest lithology, P05	1.53	1.38	1.28
Generic material, P50	1.93	1.67	1.55
Generic material, P05	1.54	1.38	1.28

Note: Slide2 9.043 with circular search; purpose-written Bishop code within 0.010. Monte Carlo for 700 m (400 samples of the dump-average envelope): median 1.67, P10 1.56, minimum 1.46. Generic material: moderate rock quality, e = 0.45, size shift from 150 to 1,000 mm.

Table 5. Factor of safety of a 700 m dump on a 10° foundation slope with a 30 m basal layer (Morgenstern–Price method)

Basal layer condition	P50 envelope	P05 envelope
Drained, dry	1.65	1.50
Drained, saturated, excess pore-pressure ratio 0.2	1.53	1.39
Undrained, $S_u = 0.17\sigma_v'$	0.65	—
Undrained, $S_u = 0.09\sigma_v'$ (liquefied)	0.39	—

Note: Nonlinear search optimized by simulated annealing in Slide2 9.043. For undrained cases the search was restricted to global mechanisms; the critical surface entered at the limit of the entry range, so the values are upper bounds for that family.